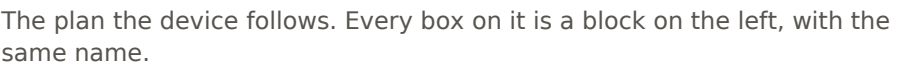
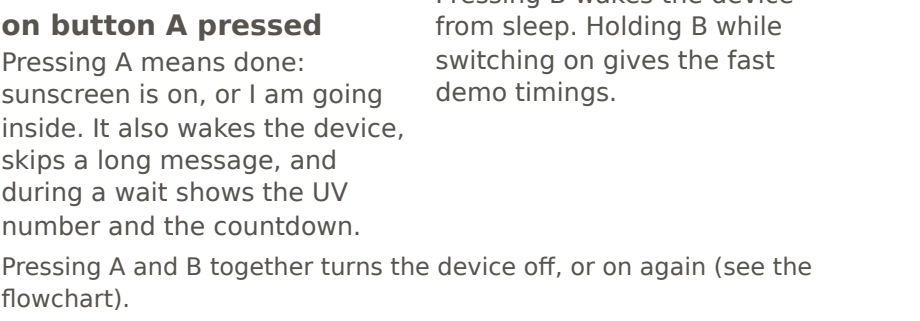


The Sunburn Flowchart, built from the real blocks of the program, with what each one does

Follow the arrows. Each red label is a box of the flowchart; the coloured picture under it is the actual block of code that does that job, exactly as it looks in the MakeCode editor, and the note says what it does. The forever block runs this plan round and round, all day.



A: done

[illegible]

All 76 blocks in the editor, in nine sections. The flowchart on this sheet is sections 1 to 4; sections 5 to 9 are the helpers: screen, sound, Bluetooth, maths and the optional screen.

Variables: remember a value	Text
Functions: a step with a name	Arrays: lists
Basic: the lights, pauses	Music: beeps
Input: buttons and time	Pins: sensor, motor, screen
Loops: repeat	Bluetooth: the phone
Logic: if, compare	Serial: the USB cable
Maths	Control: timing

UV index How strong the sun's burning rays are, from 0 (none) to 11 and above (extreme). Protection is needed from 3 upwards, even on cool or cloudy days.

micro:bit The small computer the program runs on. It has a 5 by 5 grid of lights, buttons A and B, and sockets called pins (P0, P1, P2) that other parts plug into.

Block One instruction of the program. Blocks snap together and run in order from top to bottom. A block with a name at the top (function) is a step that other blocks can call on.

Sensor The part that measures UV. It gives a voltage, which the program turns into a UV index number.

Servo A small motor with an arm. Here it taps the wearer's wrist when the sun is extreme.

Bluetooth and USB The two ways the wearable talks to a phone or a laptop. The phone app sends the online UV index; the wearable sends back what it is doing.

ms and Hz ms are milliseconds, thousandths of a second. Hz sets how high beep sounds.

Demo timings Much shorter waits (seconds instead of minutes and hours) so the whole day can be shown in a few minutes at a demo.

Names in the blocks Words like `uvSampleCount` are the program's own names for its settings and memory. Read them as English: `uv sample count` how many readings to take.

Every one of the 76 blocks, with its note, is in the booklet on the table.